Supplementary Material for
Q/R/M/C: A Stage-Decomposition Evaluation Protocol for Memory-Based Agents

Anonymous submission

Supplementary Material

This supplement is organized to support the claims of the main paper without introducing additional headline contributions. It contains the full event definitions and estimability argument, assumption stress tests, complete experimental protocols, extended tables and uncertainty estimates, scale-up results, threshold sensitivity, continuous and language-agent checks, fault-injection specifications, and reproducibility details.

Formal event definitions and estimability

For the nested regime, include the complete proof that

$$P(C^\star) = P(Q^\star)P(R^\star \mid Q^\star)P(M^\star \mid R^\star)P(C^\star \mid M^\star), \quad (1)$$

and state separately which interpretation requires the conditional stage-Markov assumption. The proof should remain short because the novelty lies in the event contract rather than in the chain rule.

Assumption stress test

Report the simulation as a sensitivity analysis of subsystem attribution. The section should identify the first diagnostic contrast that becomes unreliable as hidden between-stage state increases.

Single-agent protocol and full results

This section should contain:

- environment and task definitions;
- memory, query, candidate, and lock semantics;
- oracle-R, oracle-M, and oracle-C implementation details;
- confidence-interval and paired-test procedures;
- pooled, seed-weighted, and episode-weighted empty-retrieval estimates;
- behavior-cloning observability results;
- BabyAI portability results.

Multi-agent architecture sweep

Include separate tables for conditional rates, raw success, semantic-path completion, conditional time-to-success, and the censoring-aware metric. The text should distinguish the verified slope claim from the metric-specific $K = 8$ time optimum.

Provenance, reservation, and mechanism tests

The mechanism section should preserve the logical order used in the main paper:

1. provenance identifies the stratum in which completion is lost;
2. threshold sensitivity tests whether the stratum depends on $\varphi = 0.8$;
3. reservation retains peer evidence while changing occupancy coordination;
4. the intervention tests contention against the evidence-quality explanation.

Learned, continuous, and LLM collective checks

These experiments should be described as robustness and observability checks rather than as leaderboard comparisons.

Third-party adapter and fault-injection protocol

Report both structural and behavioral faults. Structural faults test whether the event semantics are implemented correctly. Behavioral faults test whether the aggregate stage profile contains enough signal relative to the clean control.

Comparison with diagnostic baselines

Five alternative diagnostics were evaluated on the blinded fault benchmark (63 scored cells: ten faults and a held-out control, three stacks, two models). Every method received the identical anonymized cell aggregates and the same labeled calibration control; all decision rules were fixed before scoring, with thresholds matched to the Q/R/M/C rule.

- end-to-end success summaries: accuracy 0.095, macro-F1 0.077 (no stage can be named);
- AgentBoard-style progress milestones (first marginal-rate drop): 0.730, macro-F1 0.698;

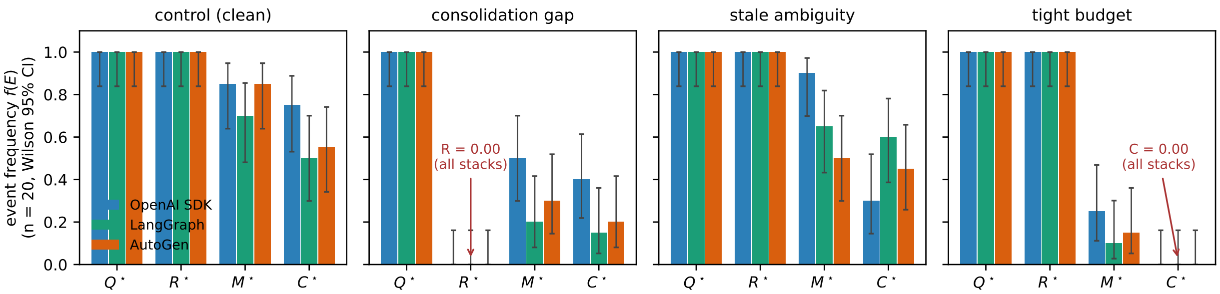

Figure 1: Marginal stage profiles for three tool-based agent stacks under a clean control, a consolidation gap, stale ambiguity, and a tight tool budget (llama3.1, $n = 20$ per cell). Structural faults produce the predicted zero-valued R or C event across all three stacks; behavioral differences are concentrated in commitment and completion.

Table 1: Marginal Q/R/M/C frequencies for third-party stacks. The events are behavioral descriptors rather than links in the chain-rule product. M_1 denotes whether the first commitment targets the correct place.

Stack	Variant	$f(Q^\star)$	$f(R^\star)$	$f(M^\star)$	$f(C^\star)$	$f(M_1)$
OpenAI SDK	control	1.00	1.00	.85	.75	.25
	consol. gap	1.00	**.00**	.50	.40	.25
	stale ambig.	1.00	1.00	.90	.30	.25
	tight budget	1.00	1.00	.25	**.00**	.25
Lang-Graph	control	1.00	1.00	.70	.50	.50
	consol. gap	1.00	**.00**	.20	.15	.10
	stale ambig.	1.00	1.00	.65	.60	.50
	tight budget	1.00	1.00	.10	**.00**	.10
Auto-Gen	control	1.00	1.00	.85	.55	.45
	consol. gap	1.00	**.00**	.30	.20	.15
	stale ambig.	1.00	1.00	.50	.45	.35
	tight budget	1.00	1.00	.15	**.00**	.15

- two-stage retrieval/execution decomposition: 0.651 at stage granularity, 0.873 at side level, macro-F1 0.450 — the deficit is concentrated where commitment must be separated from execution;
- first-deviation process conformance (order-sensitive via M_1): 0.778, macro-F1 0.755 — the deficit relative to Q/R/M/C is the budget-exhaustion branch of the conditional reading;
- supervised multinomial logistic regression on the same aggregates: 0.444 under leave-one-fault-type-out with a control false-positive rate of 1.00; 0.825 leave-one-framework-out; 0.794 leave-one-model-out; requires fault labels.

Q/R/M/C attains accuracy 0.873 (bootstrap CI over fault types $[0.71, 0.98]$), macro-F1 0.835, and control FPR $1/6$; it outperforms all exact-stage alternatives under paired McNemar tests, and all comparisons remain significant after Holm correction (adjusted p: two-stage 0.0016, progress 0.0234, conformance 0.0312, success-only and learned < 0.0001; discordant cells from 6:0 to 50:1). The learned baseline's macro-F1 under leave-one-fault-type-out is 0.435. Scored cells: $66 - 3 = 63$; the three llama3.1 cells of the prompt-only fault failed the pre-specified manipulation check (query-rate drop < 0.20) and are excluded for every method alike. Bootstrap resampling operates over fault types, not episodes, to avoid pseudoreplication. Sensitivity to the exclusion: scoring the three excluded llama3.1 cells as ground-truth "none" lowers all methods uniformly (Q/R/M/C 0.833, conformance 0.742, progress 0.697, two-stage 0.621, success-only 0.091) and leaves the ranking unchanged.

The repair study (completed 2026-07-18; llama3.1, 2,640 episodes; pre-specified in `repair_spec.json`) applies a fixed menu of stage-level repairs selected by each diagnostic. Completion gain / regret vs. the best available arm: Q/R/M/C 0.293/0.061; two-stage 0.280/0.074; conformance 0.250/0.104; progress 0.219/0.135; random 0.113/0.240; success-only 0.000/0.354; oracle best 0.354/0. Two pre-specified expectations failed: the matched repair for silent corruption recovers only 0.22–0.44 of the control gap (consolidating the true record without removing the corrupted one converts a retrieval fault into an ambiguity fault), and the budget repair applied to the clean OpenAI SDK control lowered completion by 0.15 (the budget stated in the prompt directly affects behavior). Paired bootstrap over the 27 fault–framework cells: ΔG (Q/R/M/C $-$ two-stage) $= +0.013$ $[-0.009, +0.039]$ (unresolved), vs. conformance $+0.043$ $[0.000, +0.102]$, vs. progress $+0.074$ $[+0.007, +0.157]$ (resolved). Best-arm selection share: two-stage 0.704, Q/R/M/C 0.630, conformance 0.556, progress 0.519 — coarse diagnoses are partially rescued by repair-arm overlap. Full arms matrix in `tmp/qrmc_external/repair_verdict.json`.

Threshold sensitivity and ablations

Reproducibility details